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Did FinTech Lenders Facilitate PPP Fraud?

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1 Big picture

- **Question.** Did FinTech lenders prevent or facilitate suspicious lending and possible fraud in the Paycheck Protection Program (PPP)? More broadly, what happens when online lending platforms are asked to distribute government-guaranteed credit quickly, with weak verification and little lender balance-sheet risk?
- **Setting.** The paper studies the full PPP, a COVID-19 business-relief program that distributed about **\$793 billion** through **11.47 million loans** from April 2020 to June 2021. The PPP is a useful laboratory because the loans were government-backed, often forgivable, and originated by both traditional lenders and FinTech lenders.
- **Core empirical challenge.** Fraud is not directly observed for most loans. The authors therefore cannot simply compare “fraudulent” and “legitimate” loans. Instead, they construct multiple indicators of potential misreporting and then ask whether those indicators line up across loans, lenders, time, geography, and external validation data.
- **Main contribution.** The paper is a large-scale forensic study of government credit allocation. It links PPP loan records with business registries, EIDL data, Census business data, address data, criminal-record data, crowd-sourced PPP Detective reports, and Suspicious Activity Report (SAR) aggregates.
- **Main idea.** A single suspicious feature is not proof of fraud. But if many unrelated suspicious features repeatedly point to the same loans, same lenders, same geographies, and same time periods, the evidence becomes much stronger.
- **Primary suspicious-loan indicators.** The paper uses four main flags:
 - businesses that should be registered but are missing, dissolved, or incorporated too late;
 - multiple PPP loans at the same residential address in the same draw;
 - abnormally high implied compensation relative to the borrower’s industry and location;
 - large inconsistencies between jobs reported for EIDL Advance purposes and jobs reported for PPP purposes.
- **Secondary and external validation.** The authors also study discontinuities near the \$100,000 compensation cap, rounded compensation values, excess numbers of loans in industry-county cells, clustering of loan features within lender-county cells, borrower criminal records, PPP Detective reports, and county-level SARs.
- **Main baseline result.** FinTech loans are much more likely to be suspicious. In Table II, **23.0%** of FinTech loans have at least one primary flag, compared with **7.1%** of traditional-lender loans. After rich controls and fixed effects, the FinTech excess remains economically large and statistically significant.
- **Main magnitude.** Using the four primary measures, the authors classify **1.41 million loans**, totaling **\$64.2 billion**, as suspicious. A more conservative measure requiring an additional confirming flag still identifies **950,000 loans** totaling **\$32.4 billion**. A sensitivity analysis suggests suspicious lending could be about **\$117.3 billion**.

- **Key FinTech result.** FinTech lenders originated only **32.5%** of PPP loans, but they originated **60.9%** of primary-flagged loans. The difference is not driven only by borrower type, geography, loan size, or industry composition.
- **Key time-series result.** Suspicious lending rose over the life of the program. It was about **6.3%** of loans in round 1, **7.8%** in round 2, and **16.3%** in round 3. FinTech suspicious lending rose especially sharply, reaching roughly **30%** around April and early May 2021.
- **Key incentive result.** Lenders had strong fee incentives and little credit risk. The authors estimate **\$38.0 billion** in lender processing fees, including **\$8.6 billion** to FinTech lenders. The top four FinTech lenders alone are estimated to have earned about **\$4.03 billion** in processing fees.
- **Main caveat.** The paper does not prove that every flagged loan is fraudulent. The flags can contain false positives, and some borrowers may have been eligible for smaller loans. At the same time, the flags are conservative and only apply to subsets of loans, so they likely miss many other suspicious loans.
- **Economic takeaway.** FinTech lending itself is not necessarily the problem: Square and Intuit have low suspicious-loan rates. The core problem is a combination of weak verification, limited relationships, weak reputation stakes for some lenders, high origination fees, and government guarantees that remove normal lender downside risk.

2 Data and measurement (institutional context and key objects)

2.1 PPP institutional background

- The PPP was created to provide emergency credit to businesses during COVID-19.
- Loans were generally sized as **2.5 times average monthly payroll** and could be forgiven if borrowers satisfied program requirements.
- For most borrowers, compensation above **\$100,000 per employee per year** could not be counted toward payroll for the basic loan calculation.
- Lenders originated the loans, but the government bore the credit risk and paid lenders processing fees.
- This structure created a key incentive problem: lenders could earn fees from loan volume while having limited exposure to borrower default or post-origination fraud losses.

Interpretation. The PPP is not a normal credit market. It is closer to a government-funded distribution mechanism run through private lenders. That makes lender screening incentives central.

2.2 Loan-level PPP sample

- The main data are SBA loan-level PPP data released on January 2, 2022.
- The sample includes all PPP loans originated from **April 3, 2020** through the end of the program on **June 30, 2021** that had not been repaid or canceled by January 2, 2022.
- The final sample contains **11,469,801 loans**, originated by **4,809 lenders**, with a total value of about **\$793 billion**.
- Loan-level variables include borrower name, address, business type, industry code, loan amount, reported jobs, approval date, loan draw, and lender.

Interpretation. The paper studies the universe of active PPP loans rather than a small sample. This makes it possible to detect systematic lender-level, geographic, and time-series patterns.

2.3 FinTech classification

- Lenders are classified as **FinTech** or **traditional** following Erel and Liebersohn’s methodology.
- FinTech lenders include online banks and nonbank online lending platforms.
- Classification is not always trivial because some traditional banks originated loans through online portals or partner FinTechs.
- The authors report robustness to alternative FinTech definitions.

Interpretation. The central comparison is not merely “bank versus nonbank.” It is whether the online-origination FinTech channel has systematically different suspicious-loan rates.

2.4 FinTech market share over the PPP

- FinTechs had a small initial role: about **4.2%** of loans in round 1.
- FinTech market share rose to about **20.1%** in round 2.
- In round 3, FinTech market share rose to about **46%–47%** of loans and exceeded **86%** of loans in the second half of May 2021.
- This means the FinTech channel became increasingly important as the program evolved.

Interpretation. A small problem in round 1 could become a very large fiscal problem by round 3 if suspicious lending concentrated in the rapidly growing channel.

2.5 Auxiliary data sources

The paper links PPP records to several external data sets:

- **EIDL Advance data** to compare jobs reported in another SBA program.

- **OpenCorporates business registry data** to determine whether corporation, S-corporation, and LLC borrowers appear to be legally registered and active.
- **USPS address-validation data** to identify residential, business, and central addresses.
- **U.S. Census County Business Patterns and Nonemployer Statistics** to compare compensation, receipts, and establishment counts by industry and location.
- **LexisNexis criminal-record data** for a random sample of **150,000** individual-borrower loans from rounds 1 and 2.
- **PPP Detective reports** for **147,662** loans reported by users as potentially fraudulent.
- **FinCEN SAR aggregates** to study geographic and time-series patterns in suspicious business-loan reporting.

Interpretation. The paper’s credibility comes from triangulation: suspicious PPP loans are tested against data sources that were not mechanically used to create all of the main flags.

2.6 Primary suspicious-loan flag 1: business registry

For corporate borrowers, the paper defines:

$$B_i = 1\{\text{Missing Business}_i \vee \text{Dissolved Business}_i \vee \text{Late Incorporation}_i\}. \quad (1)$$

A loan can receive this flag if the borrower is organized as a corporation, S-corporation, or LLC and:

- no matching business appears in the business registry data;
- the matched business was dissolved and inactive before the PPP loan approval date;
- the business appears to have been incorporated or initially filed after February 15, 2020, the PPP eligibility cutoff.

Interpretation. A firm that claims to be a corporation or LLC should generally have a valid registration. Missing, inactive, or too-late registrations are natural red flags.

2.7 Primary suspicious-loan flag 2: multiple loans at one residential address

The paper defines a multiple-loan flag for standardized residential addresses:

$$M_i = 1\{N_{a(i),d(i)} \geq 3 \text{ and address } a(i) \text{ is residential/noncentral}\}, \quad (2)$$

where $N_{a(i),d(i)}$ is the number of loans at the same address within the same PPP draw.

- The authors standardize addresses and remove known business or central addresses such as office buildings and apartment buildings.

- They conservatively require at least **three** loans at the same residential address within the same draw.

Interpretation. Multiple real businesses can share an address, but many loans at a modest residential property are suspicious, especially when loan amounts, industries, or reported jobs look repetitive.

2.8 Primary suspicious-loan flag 3: high implied compensation

The implied compensation per employee is approximately:

$$\widehat{Comp}_i = \frac{12}{2.5} \times \frac{LoanAmount_i}{Jobs_i}, \quad (3)$$

with adjustments for program rules and eligible business types.

The normalized compensation measure is:

$$NormComp_i = \frac{\widehat{Comp}_i}{AvgComp_{NAICS,CBSA}}, \quad (4)$$

or, for certain borrowers allowed to use gross income, the denominator is the larger of average compensation and average receipts.

The main high-compensation flag is:

$$H_i = 1\{NormComp_i > 3\}. \quad (5)$$

- Because of the \$100,000 compensation cap, this flag can only be applied in industry-CBSA cells with average compensation or receipts below \$33,333.33.
- The authors choose the threshold conservatively: more than three times the local industry benchmark.

Interpretation. A PPP loan that implies compensation far above the norm for the borrower's industry and location may reflect inflated payroll, inflated jobs, or a nonexistent business.

2.9 Primary suspicious-loan flag 4: EIDL jobs greater than PPP jobs

For 2020 EIDL Advances, the advance amount was \$1,000 per employee up to \$10,000. The paper compares the implied EIDL job count with PPP reported jobs:

$$E_i = 1\{EIDLJobs_i - PPPJobs_i \geq 3\}. \quad (6)$$

- Borrowers had an incentive to report more jobs on the EIDL Advance application because the advance amount increased with jobs.

- PPP loan size depended on payroll, not directly on the number of jobs reported, so a large EIDL–PPP discrepancy is suspicious.
- The paper uses a difference of at least **three jobs** to avoid flagging small timing or reporting differences.

Interpretation. This flag is especially useful because it compares two government programs with different incentive structures.

2.10 Combined primary and conservative measures

The main primary-flag outcome is:

$$Primary_i = 1\{B_i + M_i + H_i + E_i \geq 1\}. \quad (7)$$

The paper also considers a stricter confirmed measure:

$$Confirmed_i = 1\{Primary_i = 1 \text{ and loan } i \text{ has at least one additional primary, secondary, or external flag}\}. \quad (8)$$

Interpretation. The primary measure is broad and designed to detect many types of suspicious loans. The confirmed measure is narrower and is meant to reduce the risk that a loan is flagged because of a single honest mistake or data problem.

2.11 Secondary suspicious-loan measures

The paper uses four secondary indicators:

- **Near-\$100,000 implied compensation:** suspicious loans cluster near the PPP compensation cap.
- **Monthly rounding:** loan amounts imply round monthly compensation values, often inconsistent with detailed payroll documentation.
- **Industry-county overrepresentation:** the number of first-draw PPP loans in an industry-county cell exceeds the number of Census establishments.
- **Lender-county clustering:** loans from the same lender and county have unusually concentrated industries, loan amounts, or reported jobs.

Interpretation. These measures are not the main fraud labels. Instead, they test whether the primary flags behave like true misreporting signals.

2.12 External validation measures

The paper validates the flags using:

- felony charges in borrower criminal-record histories;

- PPP Detective crowd-sourced reports;
- county-level business-loan SAR activity during the PPP period.

Interpretation. These variables are independently informative about fraud risk. Strong alignment with the authors' flags strengthens the interpretation that the flags are not random errors.

2.13 Summary-statistics facts

- FinTech lenders originated **3.73 million loans**, or **32.54%** of loans; traditional lenders originated **7.74 million loans**, or **67.46%**.
- Average FinTech loan size was **\$23,200**, versus **\$91,211** for traditional loans. The medians were much closer: **\$18,500** and **\$20,833**.
- FinTech loans reported fewer jobs: **2.32** on average versus **10.49** for traditional loans.
- FinTech loans had higher average implied compensation: **\$62,984** versus **\$46,633**.
- FinTech borrowers were much less likely to be corporations, S-corporations, or LLCs: **19.5%** versus **65.8%**.
- FinTech round-3 borrowers were much less likely to be repeat PPP borrowers: **27.1%** versus **60.8%** for traditional lenders.

Interpretation. FinTech lending differs strongly in borrower composition. That is why the paper repeatedly uses controls, fixed effects, matched samples, and within-lender/geography designs.

3 Models and empirical specifications (all equations + interpretation)

3.1 Conceptual framework: lender incentives and suspicious lending

The paper contrasts two views of FinTech lending in the PPP.

Positive FinTech view.

- FinTech lenders may use data, automation, and online platforms to process loans quickly and reach borrowers underserved by traditional banks.
- Prior work on mortgage and small-business lending often emphasizes speed, access, and potentially better screening technology.

Fraud-facilitation view.

- Many FinTech PPP lenders had fewer preexisting borrower relationships.
- Lenders had little balance-sheet exposure because PPP loans were government-backed.

- Processing fees rewarded volume.
- Some lenders had limited reputational capital outside PPP lending.

Interpretation. The paper asks whether FinTech technology improved screening or whether the PPP incentive structure made some FinTech lenders a weak-screening channel for suspicious loans.

3.2 Baseline FinTech-versus-traditional difference

For each flag or combined flag Y_i , the paper’s adjusted comparison can be written as:

$$Y_i = \alpha + \beta FinTech_i + \Gamma X_i + \lambda_{zip(i)} + \mu_{type(i)} + \theta_{NAICS(i) \times CBSA(i)} + \varepsilon_i. \quad (9)$$

Here:

- Y_i is one of the primary flags or the combined primary flag;
- $FinTech_i$ equals one if the loan was originated by a FinTech lender;
- X_i includes loan size, number of jobs reported, and whether the loan can be matched to an EIDL Advance;
- fixed effects absorb ZIP Code, business type, and industry-by-CBSA differences.

Interpretation. The coefficient β asks: within similar ZIP Codes, business types, industries, locations, loan sizes, and job counts, are FinTech loans still more suspicious?

3.3 Unadjusted, adjusted, and matched comparisons

The paper reports three increasingly demanding comparisons:

- **Unadjusted difference:** raw difference in flag rates between FinTech and traditional lenders.
- **Adjusted difference:** regression difference after controls and fixed effects.
- **Matched difference:** each FinTech loan is matched to a traditional-lender loan with similar loan size, industry, county, and business type.

Interpretation. If the FinTech gap were purely composition, it should disappear after controls and matching. It does not.

3.4 Cross-validation among primary flags

If the primary flags are meaningful, they should be positively related to one another:

$$Flag_i^k = \alpha + \sum_{m \neq k} \rho_m Flag_i^m + \Gamma X_i + FE + \varepsilon_i. \quad (10)$$

The paper also studies odds ratios between pairs of flags.

Interpretation. Random data errors should not systematically co-occur across unrelated dimensions. Strong cross-flag correlations suggest common underlying misreporting.

3.5 Discontinuity near the \$100,000 compensation cap

PPP rules create a natural bunching point at \$100,000 of annual compensation per employee. The paper studies whether the primary flag rate jumps as implied compensation approaches the cap and falls just above it.

A stylized version is:

$$Primary_i = \alpha + f(\widehat{Comp}_i) + \tau \cdot 1\{\widehat{Comp}_i \approx 100,000\} + \varepsilon_i, \quad (11)$$

estimated separately for FinTech and traditional loans and visually using bins.

Interpretation. A borrower inflating payroll to maximize proceeds may choose values at or just below the cap. A genuine borrower with detailed payroll and benefits may legitimately appear above \$100,000, so the drop above the threshold is informative.

3.6 Rounded-loan design

The loan formula is:

$$LoanAmount_i = 2.5 \times AverageMonthlyPayroll_i + EIDLRefinance_i. \quad (12)$$

The rounding indicator is approximately:

$$Round_i = 1\{LoanAmount_i \text{ is within 50 cents of a \$1,250 interval}\}. \quad (13)$$

Interpretation. A \$1,250 loan increment corresponds to \$500 of monthly payroll. If borrowers used detailed historical payroll documents, exact round values should not be unusually suspicious. Elevated flag rates at rounded values support a misreporting interpretation.

3.7 Industry-county overrepresentation

For industry g and county c , define:

$$OverRep_{gc} = \frac{PPPFIRSTDRAWLoans_{gc}}{CensusEstablishments_{gc}}. \quad (14)$$

A secondary overrepresentation indicator is:

$$O_i = 1\{OverRep_{g(i)c(i)} > 1\}. \quad (15)$$

Interpretation. It is suspicious when PPP loans in an industry-county cell exceed the number of establishments in Census data, especially when these excess cells are concentrated in particular

FinTech lenders and also have high primary-flag rates.

3.8 Lender-county clustering

For each lender-county pair, the authors compute concentration ratios for loan industry, loan amount, and jobs reported. A generic concentration ratio is:

$$HHI_{\ell c}^m = \sum_v s_{\ell cv}^2, \quad (16)$$

where $s_{\ell cv}$ is the percentage share of loans in lender-county pair (ℓ, c) with characteristic value v .

Each concentration ratio is rescaled as:

$$\widetilde{HHI}_{\ell c}^m = \frac{HHI_{\ell c}^m - \text{Median}(HHI^m)}{P75(HHI^m) - P25(HHI^m)} \times 300 + 1000. \quad (17)$$

The overall concentration measure averages the rescaled measures:

$$\text{Conc}_{\ell c} = \frac{1}{3} \left(\widetilde{HHI}_{\ell c}^{\text{industry}} + \widetilde{HHI}_{\ell c}^{\text{amount}} + \widetilde{HHI}_{\ell c}^{\text{jobs}} \right). \quad (18)$$

Interpretation. Fraud networks may repeat the same industries, loan amounts, and job counts. Strong clustering in lender-county pairs, especially when correlated with primary flags, supports the idea of repeated questionable applications.

3.9 Secondary and external validation regression

Table III estimates regressions of the form:

$$\text{Primary}_i = \alpha + \delta S_i + \phi(S_i \times \text{FinTech}_i) + \Gamma X_i + FE + \varepsilon_i, \quad (19)$$

where S_i is one secondary or external validation measure.

Interpretation. The coefficient δ measures the relation between the validation measure and primary flags for traditional loans. The interaction ϕ asks whether the relation is even stronger for FinTech loans.

3.10 Persistence across PPP rounds

To test whether lenders learned and improved screening, Table IV estimates:

$$\text{Flag}_{i,R3} = \alpha + \beta_F \text{PastFlagPct}_{\ell,R1R2} \times \text{FinTech}_i + \beta_T \text{PastFlagPct}_{\ell,R1R2} \times \text{Traditional}_i + \Gamma X_i + FE + \varepsilon_i. \quad (20)$$

Interpretation. If lenders learned from earlier suspicious loans, β_F should be negative or small. Instead, it is positive for FinTech lenders, implying persistence and growth of suspicious lending.

3.11 Second-draw lending to previously flagged borrowers

Table V estimates:

$$SameLenderSecondDraw_i = \alpha + \beta_F FirstDrawFlagged_i \times FinTech_i + \beta_T FirstDrawFlagged_i \times Traditional_i + \Gamma X_i \quad (21)$$

Interpretation. A lender that learns from suspicious first-draw loans should be less likely to provide a second-draw loan to the same flagged borrower. Traditional lenders show that pattern more clearly than FinTech lenders.

3.12 Quantifying suspicious lending

The paper’s baseline aggregate is:

$$SuspiciousDollars = \sum_i LoanAmount_i \cdot Primary_i. \quad (22)$$

It also computes a confirmed measure:

$$ConfirmedDollars = \sum_i LoanAmount_i \cdot Confirmed_i. \quad (23)$$

Finally, it performs sensitivity analysis by lowering the high-compensation threshold from 3.0 to 2.5 times the local industry benchmark and by adding excess industry-county loans not already captured by primary flags.

Interpretation. The baseline dollar estimate is not a precise fraud number. It is a systematic estimate of suspicious lending under conservative, observable criteria.

3.13 Standard errors and inference

- Many regressions use robust standard errors double-clustered by **ZIP Code** and **lender**.
- Matched-loan differences use four-way clustering by both matched loans’ ZIP Codes and lenders.
- The clustering choices reflect two important dependence structures: suspicious lending may be geographically clustered and lender-policy clustered.

Interpretation. The paper does not treat millions of loans as independent observations. It allows for correlated behavior within lenders and places.

4 Identification and empirical design (what makes the estimates credible)

4.1 What identifies the FinTech difference

The basic FinTech-versus-traditional comparison is identified by differences in suspicious-loan indicators between lender types. The key empirical concern is composition: FinTech borrowers differ from traditional-bank borrowers.

The paper addresses this in several ways:

- controlling for loan size and jobs reported;
- using ZIP Code fixed effects;
- using business-type fixed effects;
- using industry-by-CBSA fixed effects;
- matching FinTech loans to traditional loans on loan size, industry, county, and business type;
- studying residential addresses with multiple loans;
- studying within-lender and within-geography persistence.

Identification message. The paper does not rely on one raw comparison. The FinTech gap remains after many controls that absorb observable borrower, industry, and local-area differences.

4.2 Why a single flag is not enough

Each primary flag has possible innocent explanations:

- business registries can contain errors or matching failures;
- multiple businesses can share an address;
- some workers earn high compensation relative to local industry averages;
- EIDL and PPP job counts may differ because of timing or reporting conventions.

Identification message. The paper's stronger claim comes from the fact that these flags validate one another and are also correlated with secondary and external measures.

4.3 Why cross-flag correlation matters

If the flags were merely noise, they should not repeatedly identify the same lenders and loans. Instead:

- FinTech lenders are near the top across separate flags;
- loans with high implied compensation are much more likely to have other flags;
- the high-compensation and EIDL-jobs flags have a very large odds-ratio relation;
- the same high-suspicion lenders appear in lender-level scatterplots.

Identification message. Independent suspicious features reinforce one another. This is the paper’s main substitute for direct fraud labels.

4.4 Why discontinuities and rounding strengthen the fraud interpretation

The \$100,000 cap and the loan formula create sharp program-rule features. The paper shows that suspicious flags are concentrated near these rule-driven thresholds.

- FinTech suspicious-loan rates increase sharply as implied compensation approaches \$100,000.
- Flag rates drop above \$100,000, where detailed payroll and benefit rules are more likely to matter.
- Rounded loan amounts are more likely to be flagged, especially for FinTech lenders.

Identification message. Rule-based bunching is difficult to explain with ordinary borrower composition alone. It is consistent with borrowers or intermediaries choosing simple values that maximize proceeds.

4.5 Why overrepresentation and clustering matter

The industry-county and lender-county tests are useful because they look for aggregate patterns that individual honest mistakes should not create.

- Many FinTech loans are in industry-county cells where PPP loans exceed Census establishments.
- FinTech lender-county cells show repeated industries, loan amounts, and job numbers.
- These same cells have higher primary-flag rates.

Identification message. Excess density and repeated application features suggest systematic behavior, possibly by fraud networks or weak screening pipelines.

4.6 Why external validation is important

The external validation measures are not just alternative definitions of the same flags.

- Criminal-record data provide borrower-level risk information.
- PPP Detective provides crowd-sourced qualitative allegations.
- SARs provide independent financial-institution reporting patterns.

Identification message. The primary flags line up with outside signals of suspicious activity, which reduces the probability that the results are artifacts of the authors’ construction choices.

4.7 Why the time-series evidence matters

A benign interpretation would be that FinTechs initially struggled but learned to screen better over time. The paper finds the opposite.

- Suspicious lending rises sharply in round 3.
- Lenders with high suspicious rates in rounds 1–2 continue to have high suspicious rates in round 3.
- FinTech lenders are not significantly less likely to make second-draw loans to borrowers whose first-draw loans were flagged.

Identification message. The time-series evidence is more consistent with weak incentives and learning by suspicious borrowers than with rapid lender improvement.

4.8 Main limitations

- The flags are not legal determinations of fraud.
- Some flagged loans may reflect data errors, matching errors, or legitimate unusual businesses.
- Some unflagged loans may still be fraudulent because the flags are conservative and not applicable to all loans.
- The PPP is a special setting with government guarantees and forgiveness, so the results may not generalize to ordinary FinTech credit markets.

Interpretation. The correct reading is not “every FinTech loan was bad.” The correct reading is that suspicious lending was systematically higher in certain FinTech channels under PPP incentives.

4.9 Relation to the finance literature

- The paper contributes to the FinTech literature by showing a downside of fast online credit origination when incentives are weak.
- It contributes to the fraud and misreporting literature by using forensic data methods to detect possible abuse at scale.
- It contributes to the government-program design literature by showing how weak verification and private origination incentives can create fiscal waste.
- It also echoes mortgage-crisis research showing that weakly capitalized or reputation-light originators can generate high rates of questionable lending when they do not bear full losses.

5 Tables and figures

5.1 Table I: Summary statistics

Read.

- FinTech lenders originated **3.73 million** PPP loans, compared with **7.74 million** by traditional lenders.
- FinTech loans were much smaller on average: **\$23,200** versus **\$91,211**.
- Median loan sizes were similar: **\$18,500** for FinTech and **\$20,833** for traditional loans.
- FinTech loans reported fewer jobs and higher implied compensation.
- FinTech lending became much more important in round 3.

Economic message. FinTech lenders made many small loans to different types of borrowers. Because composition differs, raw comparisons are informative but not sufficient; adjusted and matched tests are necessary.

Table I
Summary Statistics

This table presents summary statistics for our sample. The sample includes all PPP loans approved from the start of the program on April 3, 2020 through its end on June 30, 2021 that had not been repaid or canceled as of January 2, 2022. FinTech lenders are determined following Erel and Liebersohn (2022). *Loan Amount* is the initial approved amount minus any portion used to refinance an EIDL loan. *Implied Comp.* is determined following the guidelines in place when the loan was approved and is based on loan amount and jobs reported. *CBSA/NAICS Avg. Comp.* is the average compensation in the loan's industry-CBSA based on the U.S. Census CBP data. *CBSA/NAICS Avg. Receipts* is the average receipts (for business types that were able to use gross income to calculate loan size) to nonemployer businesses in the loan's industry-CBSA based on the U.S. Census NES data. *Normalized Comp.* is the ratio of *Implied Comp.* and either *CBSA/NAICS Avg. Comp.* or, if the business was able to use gross income to calculate its loan amount, the larger of *CBSA/NAICS Avg. Comp.* and *CBSA/NAICS Avg. Receipts*. *Loans (Within Draw)* at Address is the number of loans (within the loan's draw) at the same residential address. *Frac. Corp, S Corp, LLC* is the percentage of loans to these business types. *Frac. Second Draw* is the percentage of round 3 loans that are the borrower's second draw from the PPP, and *Frac. Matched EIDL Advance* is the percentage of loans with a matching EIDL Advance. *Frac. FinTech (Either Type)*, *Frac. Nonbank FinTech*, and *Frac. Online Bank FinTech* are the percentages of loans that are originated by the given type of lender.

	FinTech			Traditional		
	Mean	SD	Median	Mean	SD	Median
Num. Loans [Pct. Loans]	3,732,133 [32.54%]			7,737,668 [67.46%]		
Loan Amount	23,200	84,470	18,500	91,211	305,665	20,833
Jobs Reported	2,322	9,000	1,000	10,493	29,547	3,000
Implied Comp.	62,984	37,357	67,594	46,633	46,321	38,743
CBSA/NAICS Avg. Comp	46,921	38,895	36,942	49,798	37,459	42,849
CBSA/NAICS Avg. Receipts	46,141	31,432	35,732	59,458	37,254	53,291
Normalized Comp.	1.676	1.377	1.227	1.091	1.221	0.897
Num. Loans (Within Draw) at Address	1.321	0.794	1.000	1.193	0.801	1.000
Frac. Corp, S Corp, LLC	0.195			0.658		
Frac. Second Draw (Round 3 Loans)	0.271			0.608		
Frac. Matched EIDL Advance	0.182			0.269		
	Round 1		Round 2		Round 3	
Num. Loans [Pct. Loans]	1,618,989 [14.1%]		3,517,397 [30.7%]		6,333,415 [55.2%]	
Loan Amount	197,351		57,424		42,766	
Jobs Reported	20,468		7,915		4,561	
Implied Comp.	47,173		43,364		57,945	
CBSA/NAICS Avg. Comp	49,339		51,830		46,935	
CBSA/NAICS Avg. Receipts	—		—		50,003	
Normalized Comp.	1.156		1.040		1.477	
Num. Loans (Within Draw) at Address	1.277		1.212		1.261	
Frac. FinTech (Either Type)	0.0424		0.201		0.467	
Frac. Nonbank FinTech	0.0265		0.0789		0.410	
Frac. Online Bank FinTech	0.0159		0.122		0.0574	
Frac. Corp, S Corp, LLC	0.829		0.657		0.342	
Frac. Second Draw	—		—		0.451	
Frac. Matched EIDL Advance	0.292		0.301		0.195	

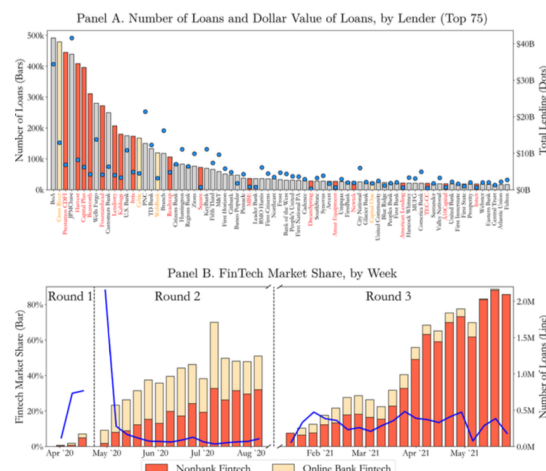


Figure 1. FinTech market share. This figure shows the role that FinTech lenders played in the PPP. Panel A shows the number of loans (bars) and dollar value of loans (dots) originated by the top 75 lenders (by number of loans). Panel B shows the percentage of loans originated by FinTech lenders during each week of rounds 1, 2, and 3 of the PPP on the left axis and the total number of loans originated each week on the right axis. In both panels, red represents nonbank FinTech lenders, cream represents online bank FinTech lenders, and gray represents traditional lenders. Note that mid-August through December 2020 is not shown in Panel B since no PPP loans were originated during this period. (Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1111/jlcr.12400))

5.2 Figure 1: FinTech market share

Read.

- Six of the top 10 PPP lenders by number of loans are FinTechs.

- FinTech market share rises from very low levels in the first week of round 1 to more than half of loans late in round 2 and more than 86% of loans in the second half of May 2021.
- Traditional banks still account for more dollar volume because their loans are larger.

Economic message. FinTech lenders became a dominant distribution channel late in the program. Any screening weakness in that channel has large aggregate implications.

5.3 Figure 2: Misreporting rates by lender

Read.

- The top 10 lenders by primary suspicious-loan rates are all FinTech lenders.
- The overall average primary flag rate is about **12.3%**.
- Some large FinTech lenders have primary flag rates above **25%**; examples include Lendistry, Capital Plus, Prestamos, Itria, and MBE.
- Not all FinTechs look bad: Capital One, Intuit, and Square have below-average suspicious-loan rates.

Economic message. The issue is not online lending per se. It is concentrated in particular FinTech lending models and incentive environments.

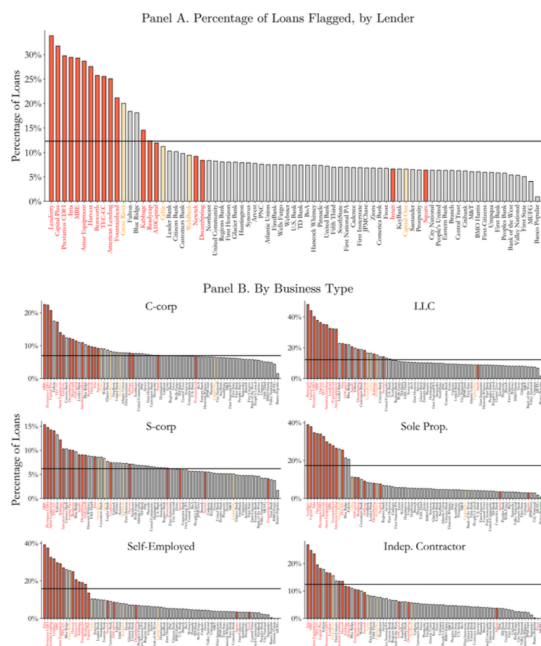


Figure 2. Misreporting rates. This figure shows the variation in the percentage of loans flagged across lenders. In both panels, the bars show the percentage of loans originated by each lender that are flagged by at least one of the primary flags and the horizontal line represents the percentage of loans originated by any lender that are flagged by at least one primary flag. The top 75 lenders by number of loans (across the entire sample) are shown. Panel A is based on all loans by each lender and Panel B is based on loans by each lender to specific business types (as denoted by each subplot title). (Color figure can be viewed at wileyonlinelibrary.com)

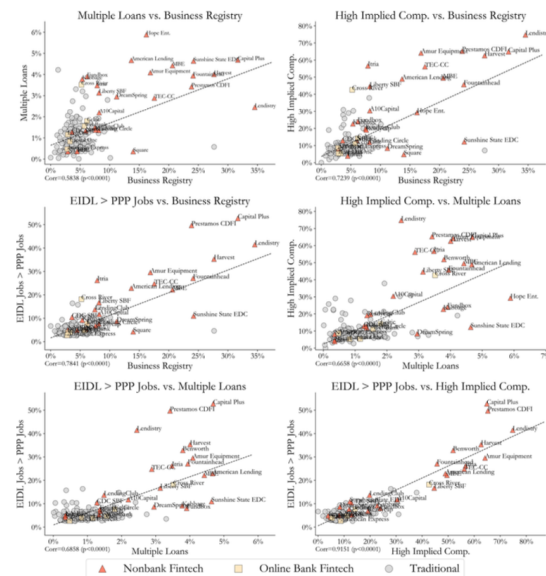


Figure 3. Relation between primary flags. This figure shows the relation between the primary flags (business registry, multiple loans, high implied compensation, or EIDL > PPP jobs flags) at the lender level. Each subpanel is a scatterplot with the percentage of loans flagged by one of the flags on each axis. Loans are filtered to the sets for which we can determine each flag for each axis separately (i.e., we do not require that both flags can be determined for a given loan). Lenders with at least 5,000 loans are shown; for the subpanels with the EIDL > PPP jobs flag, we additionally require that the lender have at least 1,000 loans with a matched EIDL Advance. The dashed line is a linear fit and the correlation is shown in the bottom left corner of each subplot. (Color figure can be viewed at wileyonlinelibrary.com)

5.4 Figure 3: Relation between primary flags

Read.

- Lender-level rates for the four primary flags are positively correlated with one another.
- The same FinTech lenders tend to appear near the top across different suspicious-loan indicators.
- No traditional lender is among the top 10 for more than one primary flag.

Economic message. The flags are not isolated measurement quirks. They consistently identify a similar set of high-suspicion lenders.

5.5 Table II: Prevalence of flags by lender type

Read.

- Business-registry flag: **10.3%** for FinTech versus **4.29%** for traditional lenders.
- Multiple-loan flag: **3.30%** versus **0.961%**.
- High implied compensation flag: **47.4%** versus **8.57%** among loans where the flag can be determined.
- EIDL > PPP jobs flag: **21.3%** versus **4.73%** among matched EIDL loans.
- At least one primary flag: **23.0%** for FinTech versus **7.13%** for traditional loans.
- After controls, the FinTech excess in the at-least-one-primary-flag measure remains **6.02 percentage points**.

Economic message. FinTech loans are substantially more suspicious even within comparable locations, industries, business types, loan sizes, and reported jobs.

5.6 Figure 4: Additional features of primary flags

Read.

- The most common EIDL–PPP job discrepancy for FinTech loans is **nine jobs**: the borrower appears to report one PPP job but enough EIDL jobs to obtain the maximum \$10,000 EIDL Advance.
- This nine-job discrepancy occurs about **14.3%** of the time for FinTech matched loans, compared with **0.5%** for traditional matched loans.
- FinTech loans have a much fatter right tail of normalized compensation: **16.4%** exceed three times local industry compensation, compared with **2.5%** of traditional loans.
- Among FinTech loans, other flag rates rise sharply with normalized compensation.

Economic message. The flags behave in the way one would expect if borrowers inflated payroll or jobs to obtain larger forgivable funds.

Table II
Prevalence of Flags by Lender Type

This table presents the percentage of loans flagged by the four main flags, at least one of the four flags, and at least two of the four flags. In Panel A, column (1) shows the percentage of FinTech loans with the given flag; column (2) shows the percentage of traditional loans with the given flag; column (3) shows the difference between the FinTech and traditional percentages; column (4) shows the adjusted differences with ZIP Code, business type, and NAICS $\times$ CBSA fixed effects and controlling for number of jobs reported, loan size, and whether we can match the loan to an EIDL Advance; and column (5) shows the differences between matched pairs of FinTech and traditional loans. The N values show the number of loans for which the flag can be determined and robust standard errors are double clustered by ZIP Code and lender. For the matched differences, robust standard errors are four-way clustered by the ZIP Code and lender of both matched loans. The full regression results for the unadjusted and adjusted differences are reported in Panels A and B, respectively, of Table IA.1 in the Internet Appendix. t -statistics are in parentheses. Significance levels are indicated by *** $p < 0.010$.

	(1)	(2)	(3)	(4)	(5)
	FinTech	Traditional	Unadjusted Difference	Adjusted Difference	Matched Difference
Business Registry	0.103 $N = 697,283$	0.0429 $N = 4,838,905$	0.0605*** (3.12)	0.0372*** (3.76)	0.0296*** (3.19)
Multiple Loans	0.0330 $N = 3,723,133$	0.00961 $N = 7,737,668$	0.0234*** (9.96)	0.0111*** (6.33)	0.0199*** (5.29)
High Implied Comp.	0.474 $N = 1,285,292$	0.0857 $N = 2,028,067$	0.388*** (8.14)	0.100*** (7.97)	0.134*** (5.35)
EIDL > PPP Jobs	0.213 $N = 677,896$	0.0473 $N = 2,084,437$	0.166*** (4.57)	0.0672*** (4.22)	0.0956*** (4.52)
At Least One Primary Flag	0.230 $N = 3,723,133$	0.0713 $N = 7,737,668$	0.159*** (8.62)	0.0602*** (8.25)	0.0754*** (7.95)
At Least Two Primary Flags	0.0239 $N = 3,723,133$	0.00329 $N = 7,737,668$	0.0206*** (8.00)	0.00744*** (5.76)	0.0137*** (7.92)

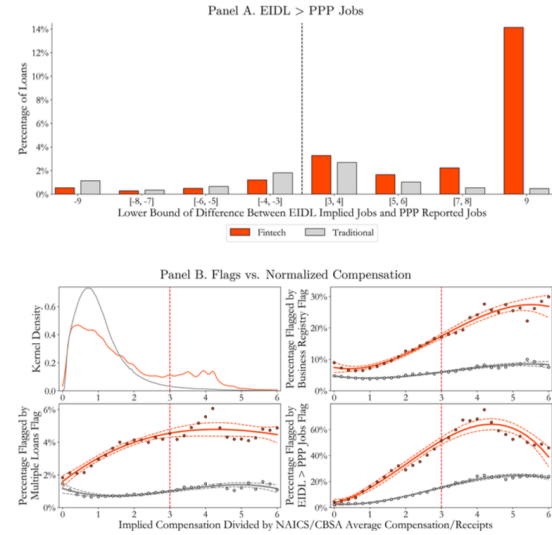


Figure 4. Additional features of primary flags. This figure shows additional features of the primary flags. Panel A shows the difference between the number of employees implied by a business's EIDL Advance amount ("EIDL Implied Jobs") and the number of jobs reported by the business on its PPP application ("PPP Reported Jobs") by lender type. Panel B shows the distribution of normalized compensation and the relation between it and the other primary flags. We define normalized compensation as the implied compensation of the loan divided by the average compensation/receipts in the loan's industry-CBSA. In the top left subpanel of Panel B, the kernel density of loans is shown, and in the other three subpanels, the percentage of loans flagged by the given flag in each bin is shown, where each bin is 0.2 units wide. The solid lines are third-degree polynomial fits for the percentage flagged and the dashed lines are 95% confidence intervals. In both panels, only loans for which the given flag can be determined are considered. (Color figure can be viewed at wileyonlinelibrary.com)

5.7 Figure 5: Discontinuities at \$100,000 and rounding

Read.

- FinTech loans bunch near the \$100,000 implied-compensation cap.
- For FinTech loans, the primary flag rate rises from about **6.5%** at \$40,000 implied compensation to about **43.5%** near \$100,000.
- There is a sharp drop in suspicious-loan rates just above \$100,000.
- FinTech loans are more likely to be rounded: **9.7%** are rounded to \$1,250 increments, compared with **7.4%** of traditional loans.
- Rounded FinTech loans have flag rates about **2.14 percentage points** higher on average.

Economic message. Threshold behavior and round-number behavior are difficult to reconcile with careful payroll documentation and strongly support the misreporting interpretation.

5.8 Figure 6: Industry-county overrepresentation

Read.

- For FinTech lenders, **39.3%** of loans are in industry-county cells where first-draw PPP loans exceed Census establishment counts.
- For traditional lenders, the comparable number is **14.0%**.

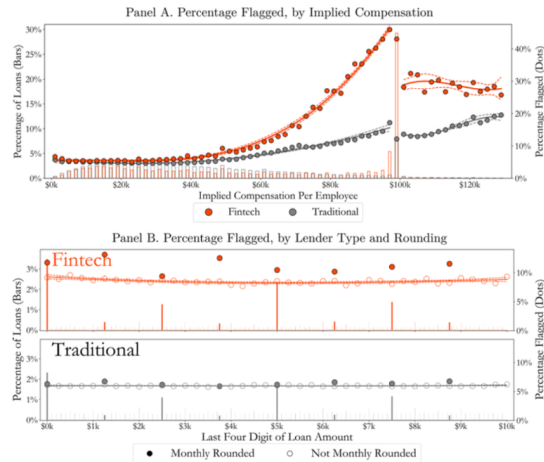


Figure 5. Discontinuities at \$100,000 and rounding. This figure shows the prevalence of the primary flags across implied compensation and loan amounts. Panel A shows the relation between implied compensation per employee and being flagged by at least one of the four main flags. Loans are binned into \$2,000-wide bins (i.e., (\$0k, \$2k], ..., (\$98k, \$100k], ..., (\$128, \$130k]). The left axis shows the percentage of loans in each bin (bars) and the right axis shows the percentage of loans in each bin that are flagged (dots). The solid lines are third-degree polynomial fits (weighted based on the number of loans in each bin), which are separately fitted for loans below \$98,000 and loans above \$100,000, and the dashed lines are 95% confidence intervals. Panel B shows the prevalence of loans being flagged by at least one of the four main flags by whether the total monthly implied compensation of a loan is rounded to an interval of \$500 (i.e., the loan amount is within ± 50 cents of an interval of \$1,250) and lender type. Specifically, the last four digits of the loan amount are considered (i.e., \$123,456.78 $\rightarrow$ \$3,456.78). The top subpanel shows FinTech loans and the bottom shows traditional loans. Further, the left axis shows the percentage of loans in each \$1-wide bin (bars for rounded compensation are thickened) and the right axis shows the percentage of loans that are flagged within each \$1-bin for monthly rounded (solid dots) and \$250-wide bin for nonrounded (hollow dots). Additionally, loans with one job reported, loans with implied compensation within $\pm 1,000$ of \$100,000, and second-draw loans to hospitality businesses are excluded. The solid lines are third-degree polynomial fits for the percentage flagged in the non-rounded bins and the dashed lines are 95% confidence intervals. (Color figure can be viewed at wileyonlinelibrary.com)

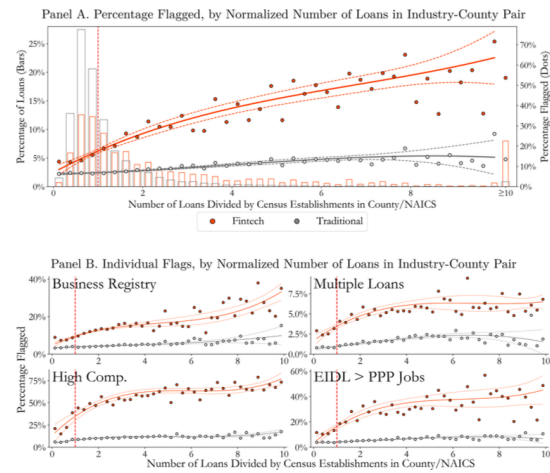


Figure 6. Overrepresentation of industries in counties. This figure shows overrepresentation of loans within industry-county pairs. We define the normalized number of loans as the number of first-draw loans divided by the number of establishments (per the 2019 U.S. Census County Business Patterns data set) in an industry (represented by NAICS code) and county pair. Panel A shows the relation between the normalized number of loans and our four main flags combined together as at least one flag and Panel B shows the relation for each flag separately. Since the CBP does not include self-employed and independent contractors as establishments, we exclude loans to these business types. Note that 6.20% of FinTech and 0.72% of traditional loans are in industry-county pairs with ratios of at least 10; these loans are represented in Panel A by the bars and dots at the far right labeled " ≥ 10 ." In both panels, loans are binned into 0.25-unit-wide bins. The solid lines are third-degree polynomial fits for the percentage of flagged loans and the dashed lines are 95% confidence intervals. (Color figure can be viewed at wileyonlinelibrary.com)

- FinTech loans exceeding establishment counts by more than a factor of two represent **32.6%** of FinTech loans, versus **8.2%** for traditional loans.
- When the loan-to-establishment ratio exceeds two, the FinTech primary flag rate reaches **41.9%**.

Economic message. The suspicious-loan problem is not only loan-level. It also appears as abnormal density of loans in specific industries and places.

5.9 Figure 7: Lender-county clustering

Read.

- High concentration ratios are much more common for FinTech loans.
- **88.6%** of FinTech loans are in lender-county pairs with concentration above 1,000, compared with **20.3%** of traditional loans.
- For FinTech loans, the flag rate rises from **10.2%** when concentration is below 1,000 to **35.7%** when concentration is above 1,300.

Economic message. Repeated loan features within lender-county cells are consistent with weak screening or repeated suspicious application pipelines.

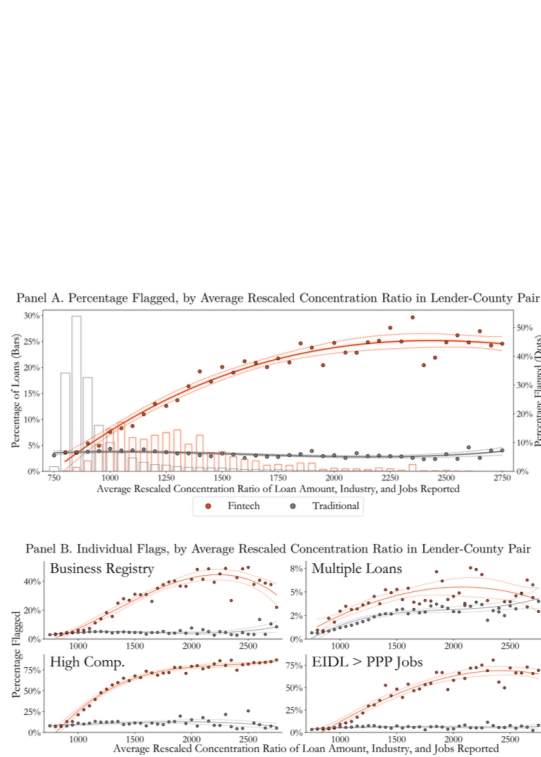


Figure 7. Clustering within lenders and counties. This figure shows clustering of loans within lender-county pairs. We calculate the concentration ratios of industries, loan amount (rounded to \$100), and jobs reported (excluding 1) for first-draw loans in each lender-county pair, rescale each concentration ratio to a median of 1,000 and IQR (interquartile range) of 300, and then take the average of the three rescaled concentration ratios. For example, let $i = 1, 2, \dots, n$ represent the n industries in a given lender-county pair, then $Concentration_{industry} = \sum_{i=1}^n s_i^2$, where s_i is the percentage of loans in the lender-county pair that are in industry i times 100 (e.g., 6.2 for 6.2%). Then, $Rescaled\ Concentration_{industry} = \frac{Concentration_{industry} - Median(Concentration_{industry})}{75^{th}Percentile(Concentration_{industry}) - 25^{th}Percentile(Concentration_{industry})} \times 300 + 1,000$. Panel A shows the relation between the average rescaled concentration ratio and our four main flags combined together as at least one flag and Panel B shows the relation for each flag separately. In both panels, only lender-county pairs with at least 25 loans are considered. Note that 2.4% of FinTech loans and 0.5% of traditional loans are outside the range of the average rescaled concentration ratio shown in Panel A. In both panels, loans are binned into 50-unit-wide bins; in Panel B, bins with fewer than 100 loans for which the given flag can be determined are excluded. The solid lines are third-degree polynomial fits for the percentage of flagged loans and the dashed lines are 95% confidence intervals. (Color figure can be viewed at wileyonlinelibrary.com)

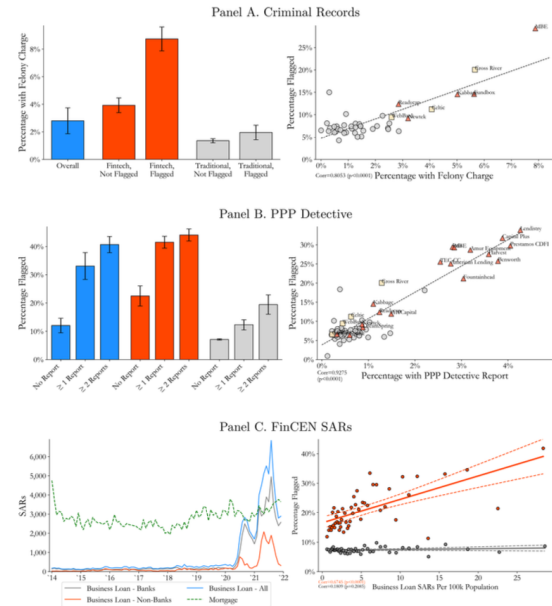


Figure 8. External validation. This figure shows relations between the external validation measures and whether the loan is flagged by at least one primary flag. Panel A shows the relation between the borrower having a felony charge from between 2000 and 2020 on their record and their loan being flagged. This is based on a sample of 150,000 round 1 and 2 loans to self-employed individuals, independent contractors, and sole-proprietors. Panel B shows the relation between the loan being reported on PPP Detective and the loan being flagged. The left panel of Panel C shows the number of SARs from banks and other financial institutions to FinCEN per month from 2014 to 2021, and the right subpanel shows the relation between the percentage flagged (separately for FinTech and traditional borrowers) and the average number of business loan SARs in 2020 and 2021 per 100,000 population (truncated at the 95th percentile) across counties. In the left subpanels of Panels A and B, the error bars denote 95% confidence intervals (standard errors double-clustered by ZIP Code and lender). In the right subpanels of Panels A and B, lenders with at least 0.2% of the sample and the top 75 lenders by number of loans, respectively, are shown. In all right subpanels, the lines are linear fits and correlations are shown in the bottom left. In the right subpanel of Panel C, the dashed lines are 95% confidence intervals. (Color figure can be viewed at wileyonlinelibrary.com)

5.10 Figure 8: External validation

Read.

- Felony charges are present for **4.7%** of sampled FinTech borrowers, compared with **1.4%** of sampled traditional-lender borrowers.
- A FinTech loan with at least one primary flag is **2.2 times** as likely to involve a borrower with a felony charge.
- PPP Detective reports are strongly related to the authors' flags: loans with no PPP Detective report have a **12.0%** flag rate, versus **33.1%** for loans with at least one report and **40.7%** for loans with two or more reports.
- The lender-level correlation between PPP Detective report rates and primary flag rates is about **0.9275**.
- Business-loan SARs spike after PPP starts and geographically align with FinTech suspicious-loan rates.

Economic message. Independent outside measures point in the same direction as the paper's constructed flags.

5.11 Table III: Secondary flags and external validation

Read.

- The dependent variable is whether a loan has at least one primary flag.
- Each secondary or external measure is interacted with a FinTech indicator.
- All FinTech interactions are positive and statistically significant.
- The largest interaction is near-\$100,000 implied compensation: the FinTech interaction is about **0.123**.
- PPP Detective, criminal records, overrepresentation, concentration, and rounding all have stronger relations with primary flags for FinTech loans.

Economic message. The paper's primary flags are systematically confirmed by secondary and external signals, especially among FinTech loans.

5.12 Figure 9: Overall suspicious lending

Read.

- The baseline primary-flag measure identifies **1,410,193 suspicious loans** totaling **\$64.2 billion**.
- FinTech lenders account for **858,820** suspicious loans totaling **\$19.4 billion**.

Table III
Secondary Flags and External Validation

In this table, we examine the relation between our four main flags, which we combine to form *At Least One Flag*, and the secondary flags. We estimate OLS regressions with *At Least One Flag* as the dependent variable and the six secondary/external flags as independent variables. Each specification also includes an interaction between the secondary/external flag and an indicator for whether the loan was originated by a FinTech lender. *\$100k Implied Comp./Receipts* is a dummy variable equal to 1 if the implied compensation/receipts per job is within $\pm \$1,000$ of \$100,000. *Monthly Rounding* is a dummy variable equal to 1 if the loan amount is within ± 50 cents of an interval of \$1,250. *Overrepresentation* is a dummy variable equal to 1 if the number of first-draw loans to businesses not listed as self-employed and independent contractors in a loan's industry-county pair exceeds the number of establishments in the industry-county pair according to the U.S. Census CBP data. *High Concentration* is a dummy variable equal to 1 if the average rescaled concentration ratio in the loan's lender-county pair is above the 75th percentile. *Felony Post-2000* is a dummy variable equal to 1 if the borrower has a felony charge on their criminal record from between 2000 and 2020. *PPP Detective* is a dummy variable equal to 1 if the loan has at least one report on PPP Detective. For all specifications, loans are filtered to the sets for which we can determine the secondary flag. Further, for specification (2), one-job loans and loans where $1/(\$100k \text{ Implied Comp.}) = 1$ are excluded. Fixed effects are as indicated at the bottom of each column. Robust standard errors are double clustered by ZIP Code and lender. *t*-statistics are in parentheses. Significance levels are indicated by * $p < 0.10$, ** $p < 0.05$, *** $p < 0.010$.

Dep. Variable: <i>1(At Least One Primary Flag)</i>						
	(1)	(2)	(3)	(4)	(5)	(6)
<i>1(\$100k Implied Comp.)</i>	0.0192** (2.45)					
<i>1(Monthly Rounding)</i>		0.00274*** (4.27)				
<i>1(Overrepresentation)</i>			-0.00609** (-1.98)			
<i>1(High Concentration)</i>				0.0116*** (3.68)		
<i>1(Felony Post-2000)</i>					0.0165* (1.82)	
<i>1(PPP Detective)</i>						0.0316*** (7.89)
$\times 1(\text{FinTech})$	0.123*** (16.21)	0.0103** (2.08)	0.0592*** (9.32)	0.0189*** (3.21)	0.0507*** (4.15)	0.0557*** (6.91)
<i>ln(Jobs Reported)</i>	Yes	Yes	Yes	Yes	Yes	Yes
<i>ln(Loan Amount)</i>	Yes	Yes	Yes	Yes	Yes	Yes
ZIP Code FE	Yes	Yes	Yes	Yes	No	Yes
Business Type FE	Yes	Yes	Yes	Yes	Yes	Yes
NAICS $\times$ CBSA FE	Yes	Yes	Yes	Yes	Yes	Yes
Lender FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	10,792,833	5,152,691	6,060,817	7,382,631	123,651	10,794,163
Num. Lenders	4,769	4,687	4,719	4,146	2,557	4,769
R^2	0.310	0.102	0.332	0.333	0.383	0.294
Mean of Dep. Var.	0.128	0.0663	0.137	0.136	0.105	0.128

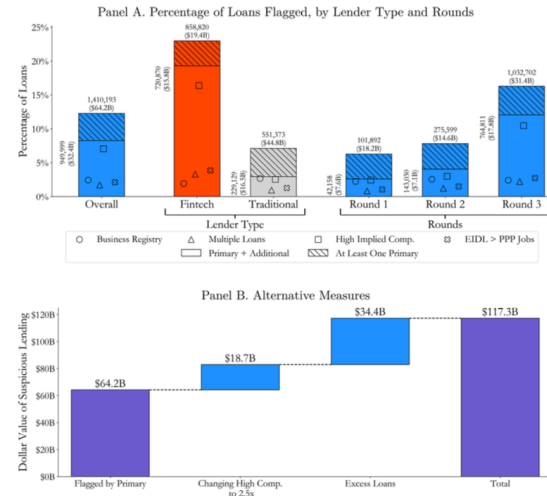


Figure 9. Overall misreporting flag rates. This figure shows the variation in the percentage of loans flagged. Panel A shows the percentage and dollar amounts of flagged loans overall, by lender type, and by round. The plain section of each bar represents the percentage of loans flagged by one primary flag and an additional flag (either another primary flag, secondary flag, or external validation) and the entire bar (plain and striped sections combined) represents loans flagged by at least one primary flag. The set of numbers to the left of each bar represent the number of loans and dollar value of loans flagged by one primary flag and an additional flag and the set on top of each bar by at least one primary flag. The markers within each bar represent the percentage of loans flagged by each of the primary flags (unconditional of whether a flag can be determined for a given loan). Panel B shows the dollar value of suspicious lending using alternative measures. The leftmost bar is the dollar value flagged by at least one primary flag, the second bar is the incremental dollar value of suspicious lending when the high compensation flag is changed from 3 to 2.5 times the industry-CBSA average compensation/receipts, the third bar is the incremental dollar value of suspicious lending based on excess loans when the loans previously flagged are considered excess first and the dollar value of any additional excess is the average of the flagged loans in the industry-county, and the fourth bar is sum of the first three bars. (Color figure can be viewed at wileyonlinelibrary.com)

- The conservative confirmed measure identifies **949,999 loans** totaling **\$32.4 billion**.
- **83.9%** of primary-flagged FinTech loans are confirmed by an additional flag, compared with **41.6%** of primary-flagged traditional loans.
- A sensitivity analysis using a 2.5-times compensation threshold and excess industry-county loans suggests suspicious lending of about **\$117.3 billion**.

Economic message. Even conservative definitions imply large fiscal exposure. Less conservative but still systematic definitions imply much larger suspicious lending.

5.13 Figure 10: Persistence and growth across rounds

Read.

- Suspicious lending grows over time, especially for FinTech lenders.
- FinTech suspicious-loan rates start below 10% but rise toward roughly **30%** near the end of the program.
- ZIP Codes with high suspicious-loan rates in rounds 1 and 2 also have high suspicious-loan rates in round 3.
- The increase is especially pronounced in FinTech lending.

Economic message. The evidence does not fit a simple story in which FinTech lenders made early mistakes and then improved. It is more consistent with persistent weak screening and growing exploitation of the program.

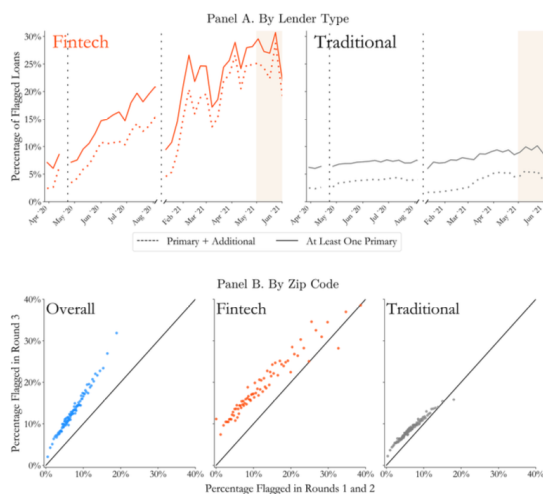


Figure 10. Persistence and growth across rounds. This figure shows the persistence and growth of flagged loans across lending rounds. Panel A shows this by lender type and Panel B by ZIP Code. In Panel A, each subpanel shows a lender type and each series is the percentage of loans flagged by the given measure across time. In Panel B, the percentage of loans flagged in rounds 1 and 2 is shown on the horizontal axis and in round 3 on the vertical axis. For Panel A, the vertical dotted lines split each subpanel into the three lending rounds. The solid lines are loans flagged by at least one primary flag and the dashed line is loans flagged by at least one primary flag and an additional flag (another primary, secondary, or external). For Panel B, the left subpanel uses all loans, the middle uses FinTech loans, and the right uses traditional loans. ZIP Codes with at least 100 loans in rounds 1 and 2 combined are shown. The black line is a 45° line. (Color figure can be viewed at wileyonlinelibrary.com)

Table IV
Persistence of Lender Behavior across Rounds

In this table, we examine the persistence of lender behavior across rounds. We estimate OLS regressions with dummies for whether each round 3 loan is flagged by our four main flags individually (specifications (1) to (4)) and at least one of them (specification (5)) as the dependent variable and the percentage of the lender's loans were flagged by the same flag in rounds 1 and 2 as the independent variable. Interactions with whether the loan was originated by a FinTech or traditional lender are included in all specifications. For specifications (1) to (4), loans are filtered to the sets for which we can determine the flag. Further, to ensure that we have accurate measures of past behavior, we require that each lender have at least 100 loans in rounds 1 and 2 (combined) for which we can determine the given flag. Fixed effects are as indicated at the bottom of each column. Robust standard errors are double clustered by ZIP Code and lender. *t*-statistics are in parentheses. Significance levels are indicated by ***p* < 0.05, ****p* < 0.010.

Dep. Variable:	(1) <i>1(Business Registry)</i>	(2) <i>1(Multiple Loans)</i>	(3) <i>1(High Implied Comp.)</i>	(4) <i>1(EIDL > PPP Jobs)</i>	(5) <i>1(At Least One Flag)</i>
<i>Past Pct. This Flag × 1(FinTech)</i>	0.905*** (4.67)	0.627*** (3.69)	0.361*** (2.72)	0.861*** (2.97)	0.456*** (4.50)
<i>× 1(Traditional)</i>	0.167 (0.73)	0.211** (2.20)	-0.453*** (-3.83)	-0.173 (-1.26)	-0.185 (-1.55)
<i>In(Jobs Reported)</i>	Yes	Yes	Yes	Yes	Yes
<i>In(Loan Amount)</i>	Yes	Yes	Yes	Yes	Yes
<i>ZIP Code FE</i>	Yes	Yes	Yes	Yes	Yes
<i>Business Type FE</i>	Yes	Yes	Yes	Yes	Yes
<i>NAICS × CBSA FE</i>	Yes	Yes	Yes	Yes	Yes
Observations	1,830,109	5,280,796	1,574,978	1,010,713	5,280,796
Num. Lenders	2,432	3,078	1,547	1,409	3,078
<i>R</i> ²	0.141	0.0512	0.630	0.346	0.366
Mean of Dep. Variable	0.0705	0.0208	0.341	0.135	0.164

5.14 Table IV: Persistence of lender behavior

Read.

- For FinTech lenders, past suspicious-loan rates in rounds 1–2 strongly predict suspicious-loan rates in round 3.
- The FinTech persistence coefficient is **0.905** for the business-registry flag, **0.627** for the multiple-loan flag, **0.361** for high implied compensation, **0.861** for EIDL > PPP jobs, and **0.456** for at least one primary flag.
- Traditional-lender coefficients are smaller, often insignificant, and sometimes negative.

Economic message. High-suspicion FinTech lenders did not appear to clean up their lending in round 3. Their suspicious behavior persisted.

5.15 Table V: Likelihood of receiving a second-draw loan

Read.

- Traditional lenders are significantly less likely to provide a second-draw loan to borrowers whose first-draw loans are flagged.
- The unconditional traditional-lender effect is about **-3.09 percentage points**.

- The comparable FinTech effect is only about **-1.02 percentage points** and is statistically insignificant.
- Conditional on receiving a second-draw loan, the traditional-lender effect remains negative and significant, while the FinTech effect is not significant.

Economic message. Traditional lenders show some evidence of avoiding previously suspicious borrowers. FinTech lenders do not show similar learning or discipline.

Table V
Likelihood of Receiving a Second-Draw Loan

In this table, we examine whether lenders were more/less likely to provide a second-draw loan to a borrower whose first-draw loan is flagged by at least one of our primary flags. We estimate OLS regressions with a dummy for whether the same lender provided the first- and second-draw loans as the dependent variable and a dummy for whether the first-draw loan was flagged by at least one of the primary flags as the independent variable. In specifications (1) and (2), if a borrower did not receive a second-draw loan, the dependent variable is set to 0, and in specifications (3) and (4), only borrowers who received both a first- and second-draw loans are included in the sample. In the even specifications, interactions for whether the first-draw loan was originated by a FinTech or a traditional lender are included. Fixed effects are as indicated at the bottom of each column. Robust standard errors are double clustered by ZIP Code and lender. *t*-Statistics are in parentheses. Significance levels are indicated by **p* < 0.10, ****p* < 0.010.

Dep. Variable: 1(First and Second Draw by Same Lender)				
	(1) Unconditional of Receiving Second Draw	(2) Unconditional of Receiving Second Draw	(3) Conditional on Receiving Second Draw	(4) Conditional on Receiving Second Draw
<i>1(First Draw Flagged)</i>	-0.0261*** (-6.54)		-0.00878* (-1.84)	
× <i>1(FinTech)</i>		-0.0102 (-0.84)		0.0122 (0.55)
× <i>1(Traditional)</i>		-0.0309*** (-11.41)		-0.0156*** (-4.86)
<i>ln(Jobs Reported)</i>	Yes	Yes	Yes	Yes
<i>ln(Loan Amount)</i>	Yes	Yes	Yes	Yes
Zip Code FE	Yes	Yes	Yes	Yes
Business Type FE	Yes	Yes	Yes	Yes
NAICS × CBSA FE	Yes	Yes	Yes	Yes
Lender FE	Yes	Yes	Yes	Yes
Observations	4,847,988	4,847,988	1,579,897	1,579,897
Num. Lenders	4,647	4,647	4,446	4,446
R ²	0.121	0.121	0.415	0.415
Mean of Dep. Var.	0.278	0.278	0.836	0.836

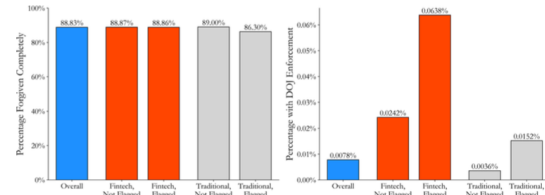


Figure 11. Enforcement and forgiveness. This figure shows the forgiveness of and DOJ enforcement actions regarding PPP loans. The left subpanel shows the percentage of round 1 loans that have been completely forgiven as of January 3, 2022. The right subpanel shows the percentage of round 1 and 2 loans that are part of DOJ enforcement actions as of March 1, 2022. In both panels, the overall percentage and the percentage by lender type and whether the loan is flagged by at least one primary measure are shown. (Color figure can be viewed at wileyonlinelibrary.com)

5.16 Figure 11: Enforcement and forgiveness

Read.

- Round-1 suspicious loans are forgiven at rates similar to nonsuspicious loans.
- DOJ enforcement actions are more common for flagged and FinTech loans, but the enforcement rate is below **0.1%** in all categories.
- As of March 1, 2022, DOJ publicized **258 criminal complaints** involving only **502 loans**, compared with the paper's **1.41 million** primary-flagged loans.
- Between the May 2021 and January 2022 SBA releases, **234,727** loans were canceled; **30.7%** of these canceled loans had at least one primary flag.

Economic message. Expected penalties appear very low relative to the scale of suspicious lending. This weak enforcement environment helps explain why suspicious lending could expand over time.

5.17 FinTech incentives and lender business models

Read.

- The paper estimates **\$38.0 billion** of PPP lender processing fees.
- FinTech lenders received about **\$8.6 billion**.
- The top four FinTech lenders are estimated to have earned about **\$4.03 billion**: Prestamos **\$1.06 billion**, Cross River **\$1.03 billion**, Harvest **\$1.02 billion**, and Capital Plus **\$938 million**.
- Some large FinTech PPP lenders had limited SBA lending histories before the pandemic.
- Square and Intuit are counterexamples: they are FinTech lenders with low suspicious-loan rates, likely reflecting stronger customer relationships, data, and reputational stakes.

Economic message. The problem is not technology alone. The problem is technology combined with weak screening incentives, weak reputational discipline, and government-funded upside.

6 Short summary

- The paper studies whether FinTech lenders facilitated suspicious PPP lending.
- It builds four primary suspicious-loan flags using business registrations, residential-address clustering, high implied compensation, and EIDL–PPP job inconsistencies.
- FinTech loans are much more likely to be flagged: **23.0%** versus **7.1%** for traditional loans.
- The FinTech gap remains after rich controls, fixed effects, and matched comparisons.
- Secondary tests show bunching near the \$100,000 cap, suspicious rounding, industry-county overrepresentation, and lender-county clustering.
- External validation from criminal records, PPP Detective, and SARs supports the interpretation that the flags capture real suspicious lending.
- The baseline estimate is **1.41 million suspicious loans** and **\$64.2 billion**; a broader sensitivity estimate is **\$117.3 billion**.
- Suspicious lending increased over time rather than declining.
- FinTech lenders earned large processing fees and bore little credit risk.
- The policy lesson is that fast lending platforms need strong verification and incentive-compatible oversight when public money is at stake.

7 Conclusion

7.1 Core conclusion

Griffin, Kruger, and Mahajan show that suspicious PPP lending was widespread and disproportionately concentrated among FinTech lenders. Their evidence is not based on one fraud proxy.

It comes from many distinct signals that repeatedly identify the same patterns: FinTech concentration, high implied compensation, residential-address clustering, EIDL discrepancies, excess industry-county loan density, repeated application features, criminal-record validation, PPP Detective reports, SARs, and weak enforcement.

7.2 Economic intuition

The PPP changed the lending problem. In normal lending, a lender that screens poorly suffers losses or reputational costs. In the PPP, lenders earned fees, the government bore credit risk, and many loans were forgivable. Under these incentives, lenders with weak relationships, limited reputation, and high-volume online origination could profit from speed even when screening quality was poor.

7.3 Takeaway

FinTech can expand credit access and process applications quickly, but speed is not automatically social value. When loans are publicly guaranteed and screening is weak, fast origination can scale fraud as well as access. The paper's central policy lesson is that government lending programs must align lender incentives, require verification, and maintain credible enforcement before funds are disbursed at massive scale.